Diatomic gas ($f = 5$, $\kappa = 1.4$)

shock angle β [°]

deflection angle θ [°]

1.05 1.1 1.15 1.2 1.25 1.3 1.35 1.4 1.45 1.5 1.6 1.7 1.8 1.9 2 2.1 2.2 2.3 2.4 2.5 2.6 2.8 3 3.2 3.4 3.6 3.8 4 4.5 5 6 8 10 15 20 ∞

$\widehat{M} = 1$

weak solution
strong solution
sonic downstream flow $\widehat{M} = 1$
curve labels: upstream Mach number M
curve tips: maximum deflection

